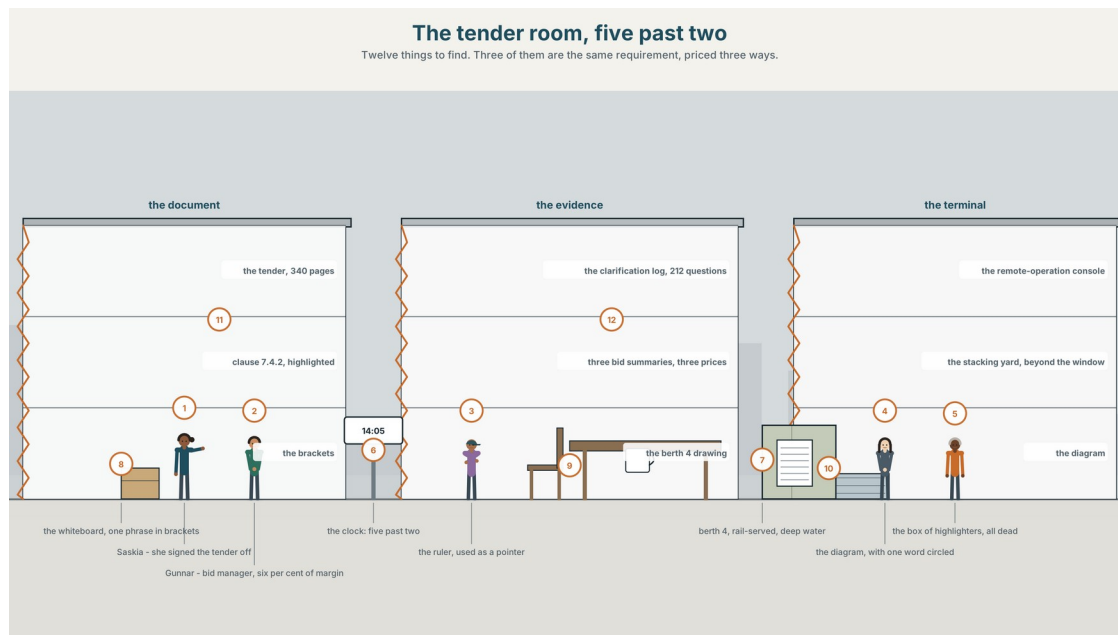


Table of Contents

B2.2 • Unit 3 • Fourteen Words, One Noun

English Animated • *Holding the Floor* • a port and its logistics chain, Rotterdam Student's Book pages 50–71 • 22 pp • ~11 hours

Part 0 • The Big Picture ♦ A • THE WORLD



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The tender room, five past two — Twelve things to find. Three of them are the same requirement, priced three ways.

How much can one noun carry?

By the end of this unit you can pack a clause into a noun phrase and take somebody else's apart. Two hundred words.

A three-hundred-and-forty-page tender, one fourteen-word noun phrase, and three suppliers who read it three ways

♦ A • THE WORLD

♦ B • THE SYSTEM

complex noun phrases, before and after the head • 28 words for systems and capacity • 22 words and phrases that stack • chunking a heavy noun phrase

◆ A • THE WORLD

A three-hundred-and-forty-page tender, one fourteen-word noun phrase, and three suppliers who read it three ways

◆ C • THE PERFORMANCE

Two hundred words dense enough for a specification and read the same way twice

0A • Find it

Find twelve things in this picture. Three of them are the same requirement, priced three different ways.

the tender document, 340 pages · clause 7.4.2, highlighted · the clarification log, 212 questions · three bid summaries, three prices · a whiteboard with one noun phrase broken into brackets · the stacking yard beyond the window · a crane remote-operation console · the berth 4 drawing · a ruler used as a pointer · a diagram with one word circled · a clock at 14.05 · a box of highlighters, all dead

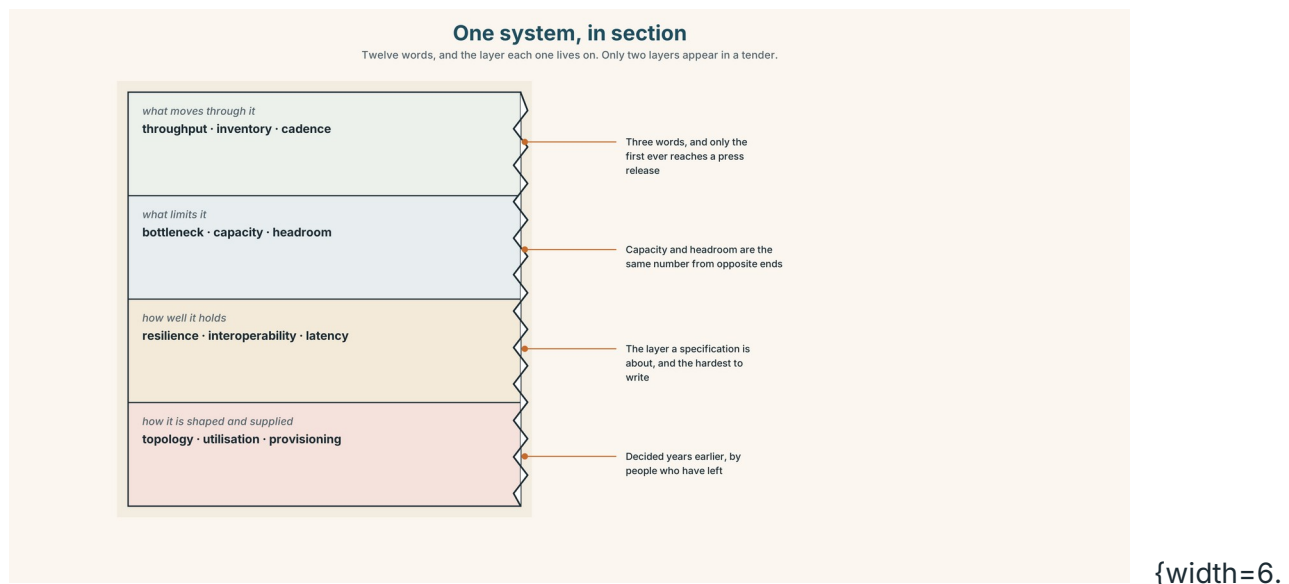
0B • Guess it

Three documents in this picture disagree by two point three million euros. Which three, and where exactly does the disagreement live?

0C • Say it

Record sixty seconds. Describe a system you use — at work, at college, anywhere — in as much detail as you can manage. You will record it again at the end of this unit.

Part 1 • Vocabulary Lab 1 ◆ B • THE SYSTEM



6in}

One system, in section — Twelve words, and the layer each one lives on. Only two layers appear in a tender.

1A • Meet — the words of a system

Match each word to one layer of the cutaway.

throughput · bottleneck · capacity · resilience · interoperability · latency · utilisation · topology · inventory · headroom · cadence · provisioning

1B • Sort — how much goes through, where it stops, or how the whole thing is built?

Three columns. **Two of the twelve are the same measurement from opposite ends: one says how much is being used and the other how much is left.**

throughput · bottleneck · resilience · capacity · latency · interoperability · utilisation · headroom · topology · inventory · cadence · provisioning

1C • Meet — the five phrases a specification turns on

a single point of failure	One thing whose loss stops everything. Every system has one.
the critical path	The chain of tasks that decides the finish date, and only that chain.
spare capacity	What you are paying for and not using, which is what resilience is.
lead time	The gap between asking and getting, and the hardest number to shorten.
a service level	A promise with a number in it. Without the number it is a hope.

1D • Collocation Lab

Complete each one, and say which of the five is the only one with a time in it.

peak demand · end to end · scheduled maintenance · turnaround time · design margin

1. The terminal is sized for _____, which occurs on about nine days a year.
2. _____ is twenty-two hours for a feeder and four days for a deep-sea call.
3. _____ takes two cranes out for six weeks every second year.
4. The clause specifies the latency _____, from console to crane and back.
5. The specification carries a twenty per cent _____, which three suppliers priced three ways.

1E • The phrasal verbs, and the three fixed expressions

scale back · build in · roll out

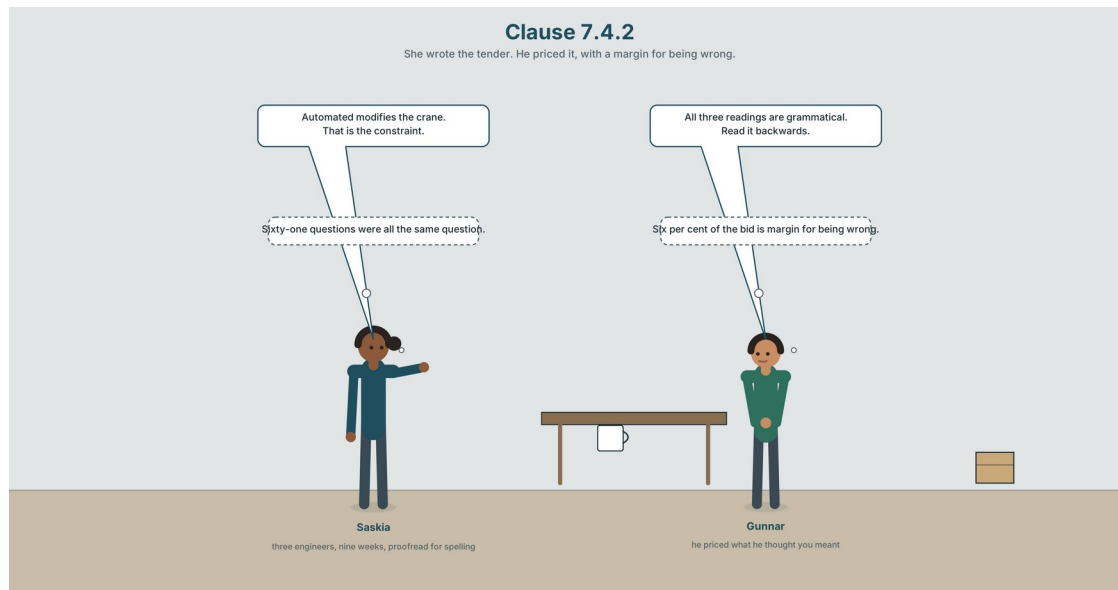
¶ it depends on the volume · ¶ that is the constraint · ¶ there is no slack

One of these fixed expressions ends an argument and two continue it. Which ends it, and what must be true for it to be fair?

1F • Own it

Write one noun phrase of at least six words describing something in your own field, and then circle the head.

Part 2 • Listening ◆ A • THE WORLD



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Clause 7.4.2 — *She wrote the tender. He priced it, with a margin for being wrong.*

2A • Clause 7.4.2 🎧 Track 3.1

Before you listen. The tender runs to three hundred and forty pages. Clause 7.4.2 contains a fourteen-word noun phrase. Three suppliers have priced it, and the prices differ by two point three million euros.

Listen once. What do the three suppliers each think the clause is asking for?

Listen again. Complete the table.

	What they think the head noun is	What they priced	The difference
supplier A			
supplier B			
supplier C			

Listen again. True or false?

- The tender is three hundred and forty pages. ☐ T ☐ F
- All three suppliers priced the same scope. ☐ T ☐ F
- The clarification log has two hundred and twelve questions. ☐ T ☐ F
- The phrase has a post-modifier at the end. ☐ T ☐ F

Noticing. Four versions of the same requirement.

- the automated gantry crane handover latency requirements for berth 4*
- the latency requirements for handover of the automated gantry cranes at berth 4*
- the requirements for handover latency on automated gantry cranes at berth 4*

- at berth 4, handover latency on automated gantry cranes shall not exceed 400 ms

All four describe one thing. Find the head noun in each of the first three, and say which of the four could be priced without a question.

2B • Decoding clinic 🎧 Track 3.2

Thirty seconds at speed. Three jobs.

- One speaker reads a fourteen-word noun phrase aloud. Mark where they pause.
- Which word carries the main stress, and why that one?
- Somebody says *read it backwards*. Do it, and say what it is for.

Part 3 • Grammar Lab 1 ♦ B • THE SYSTEM

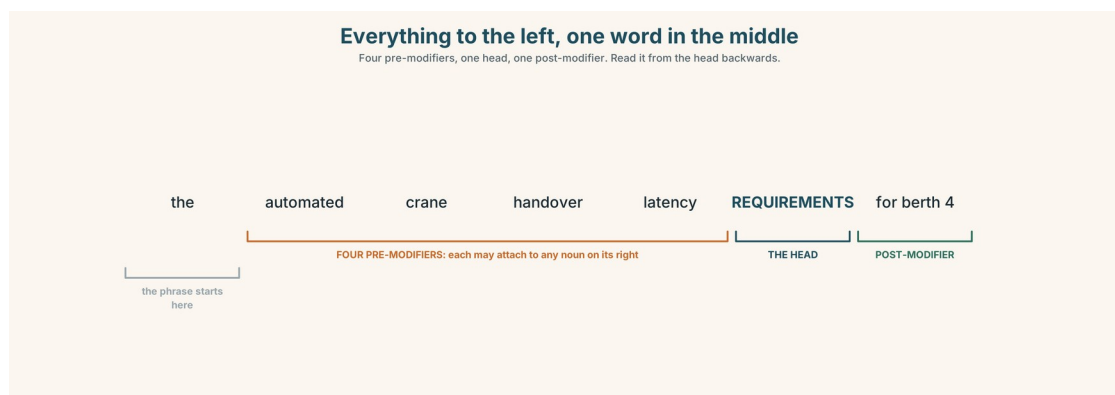
Everything to the left, and everything to the right

3A • Notice

Look at the four versions in Part 2 again.

- In version 1, which word is the head? **How did you find it?**
- Version 2 and version 3 have different heads. **Which, and what changes?**
- Version 4 has no long noun phrase at all. **What has it got instead?**
- Why can version 1 be read three ways and version 4 only one?**

3B • See



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Everything to the left, one word in the middle — Four pre-modifiers, one head, one post-modifier. Read it from the head backwards.

3C • Know

the head	the last noun before any <i>of, for, at</i> or relative clause	... <i>latency requirements for berth 4</i>
pre-modifiers	adjectives and nouns stacked to the left, no limit	<i>automated gantry crane handover latency</i>

post-modifiers	prepositional phrases, relatives, participles, to the right	<i>...for berth 4, which is rail-served</i>
the reading rule	work right to left from the head	<i>requirements ← latency ← handover ← crane</i>
the ambiguity	each pre-modifier may attach to any noun on its right	<i>automated — the crane, or the handover?</i>
the cure	one post-modifier, or one verb	<i>handover latency shall not exceed 400 ms</i>

A noun phrase is a clause with the verb taken out. *The automation of the cranes was deployed* becomes *crane automation deployment*, and three words have replaced nine. The density is the point; the cost is that nobody can tell what did what to what.

English stacks nouns to the left and English readers parse from the head. That is why a long pre-modified phrase is read backwards, and why the sixth word of a fourteen-word phrase is the one that nobody can place. Post-modification is longer and never ambiguous, which is the whole trade.

the stackable set	<i>container · handling · terminal · automation · deployment · integration · allocation · scheduling</i>
phrases they build	<i>a container handling terminal · a terminal operating system · a customs clearance delay</i>
what systems people do	<i>stack up · take up · set up</i>
how you take one apart	<i>three nouns in a row · what modifies what · read it backwards</i>

Why it matters. A specification is a document people price. A noun phrase that can be read two ways is not a style problem; it is two point three million euros, and the people who find it are the ones who read backwards.

3D • Use — controlled

Turn each clause into a noun phrase.

They deploy automation on the cranes. → (1) _____ The system schedules the allocation of berths. → (2) _____ Customs clearance was delayed. → (3) _____ They integrated the terminal operating system. → (4) _____ The waterway network inland was extended. → (5) _____

3E • Use — guided

Unpack each noun phrase into a sentence with a verb and name every participant.

1. crane automation deployment delays
2. container handling terminal capacity constraints
3. terminal operating system integration requirements
4. berth allocation scheduling cadence
5. inland waterway network provisioning headroom

3F • Use — open

In pairs. One of you says a four-noun phrase from your own field. The other reads it backwards aloud and then asks *what modifies what?* **Six rounds each.**

3G • Error autopsy

× *the automated of the gantry crane handover latency* × *the handover latency requirements which for berth 4* × *the berth 4 for handover latency requirements*

For each, name the rule and say what the writer was reaching for.

3H • Check — eight items

1. They deploy automation on the cranes. → _____ (*three words*)
2. The system schedules berth allocation. → _____ (*three words*)
3. Customs clearance was delayed. → _____ (*three words*)
4. They integrated the terminal operating system. → _____ (*five words*)
5. In *automated gantry crane handover latency requirements*, name the head.
6. × *the berth 4 for handover latency requirements* → correct it.
7. Rewrite *crane automation deployment delays* as a sentence with two participants.
8. Rewrite item 5 so that it can only be read one way.

Part 4 • Speaking ♦ C • THE PERFORMANCE

4A • Fluency starter — add one to the left

One person says a two-word noun phrase. Each person adds one modifier to the left. **The round ends when somebody cannot say what the head is** — and everybody counts the words.

4B • The functional core — packing information

The same requirement, two ways
One of them was priced three ways. The other could only be priced once.

clause 7.4.2, as issued	as rewritten
14 words • 0 verbs • 0 numbers • 3 readings	17 words • 1 verb • 1 number • 1 reading
	
Find the six. One version contains a number. Which, and how many words does it take to get to it? 6 differences — find them all	

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The same requirement, two ways — One of them was priced three ways. The other could only be priced once.

Language Bank

packing	<i>container handling · terminal operating · customs clearance · cold-chain</i>
anchoring the head	<i>requirements for · capacity at · access to · integrity of</i>
unpacking	<i>what modifies what · read it backwards · three nouns in a row</i>
pricing it	<i>it depends on the volume · that is the constraint · there is no slack</i>

Information gap. Learner A has clause 7.4.2. Learner B has the drawing of berth 4. **Agree on one sentence that a supplier could price.** Ten minutes.

4C • The performance — three minutes

Specify something in three minutes — a process, a device, a service. **Every requirement must be one sentence with one number in it,** and your partner must be able to repeat any of them back.

Record it.

4D • Observer

One person writes down every noun phrase of four words or more and marks the head in each.

4E • Three criteria

- ☐ Could my partner find the head of every phrase I used? ☐ Did each requirement have a number? ☐ Did I pack where packing helped, and unpack where it did not?

Part 5 • Reading A • THE WORLD

5A • Predict

The title is "**Two point three million euros.**" **How can three companies reading one sentence arrive at three prices?**

5B • Read — two minutes

Two point three million euros

The tender for the berth 4 automation package runs to three hundred and forty pages. Clause 7.4.2 specifies, in a single noun phrase of fourteen words, the handover latency between the remote console and the crane.

Three suppliers bid. Their prices differ by two point three million euros, and the whole of the difference sits in that clause.

Supplier A read the phrase as a requirement about the cranes, and priced new control hardware. Supplier B read it as a requirement about the handover, and priced a software layer. Supplier C read *automated* as modifying *handover* rather than *crane*, and priced neither, on the grounds that the handover is already automated.

All three readings are grammatical. Two of them are wrong, and the tender does not contain a sentence that says which.

The clarification log for this tender holds two hundred and twelve questions. Sixty-one of them are of the form *does X modify Y or Z*. The port answered all sixty-one individually, by email, to the supplier who asked, and published none of the answers.

The specification was written by three engineers over nine weeks. It was proofread for spelling.

5C • Comprehension — eight questions

- 1. How long is the tender, and what does clause 7.4.2 specify?
- 2. How much do the prices differ by?
- 3. What did each supplier price?
- 4. How many questions are in the clarification log, and how many are about modification?
- 5. How were the sixty-one answered?
- 6. **Why does the writer say "all three readings are grammatical"?**
- 7. **The answers were sent individually and not published. What does that do to a tender?**
- 8. **The last sentence is seven words long. What is it doing, and why does it come last?**

5D • Vocabulary in context

Infer from the text, then check.

sits in that clause · on the grounds that · of the form · individually, by email · proofread for spelling

5E • The counter-text

Clause 7.4.2, every modifier traced

Fourteen words, one head, and three defensible readings of the same sentence.

BERTH 4 AUTOMATION PACKAGE - CLAUSE 7.4.2

"The automated rail-mounted gantry crane remote operation handover latency requirements for berth 4 shall be met in full."

Annex C, test method: "console input to confirmed crane response, measured end to end." No figure is given in the clause itself.

SUPPLIER	WHAT THEY READ AS THE HEAD	PRICED
A	CRANE REQUIREMENTS	NEW CONTROL HARDWARE
B	HANDOVER REQUIREMENTS	A SOFTWARE LAYER
C	AUTOMATED HANDOVER	NEITHER - ALREADY DONE
SPREAD BETWEEN BIDS	EUR 2.3M	--
WORDS IN THE PHRASE	14	--
NUMBERS IN THE CLAUSE	0	--
CLARIFICATIONS, THIS TENDER	212	--
OF THOSE, ABOUT MODIFICATION	61	NONE PUBLISHED

WHAT EACH PRE-MODIFIER COULD ATTACH TO

automated
rail-mounted
remote
for berth 4

crane, operation, handover or latency
crane only - a physical fact saves it
operation or handover
requirements - post-modifiers do not stray

Three engineers wrote this over nine weeks. It was proofread for spelling, and every word in it is spelled correctly.

Fourteen words, no verb, and the head is the twelfth

The number is in an annex and the requirement is in the clause

Three readings, three scopes, and two point three million euros

Only two of the modifiers are pinned, and both by physics

Proofread for spelling is the whole sentence

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Clause 7.4.2, every modifier traced — Fourteen words, one head, and three defensible readings of the same sentence.

Read clause 7.4.2 and answer.

1. Where is the head noun, counted in words from the start?
2. **Mark every pre-modifier and say which nouns to its right it could attach to.**
3. The clause has one post-modifier. **What does it attach to, and could it attach to anything else?**
4. **Rewrite the clause as three sentences, each with one number.**

5F • Two texts

The clause says *automated gantry crane handover latency requirements for berth 4*. The drawing says *console to crane, 400 ms, berth 4 only*.

Which is the specification? Write two sentences, and use the word *interoperability* in one of them.

Part 6 • Vocabulary Lab 2 ♦ B • THE SYSTEM

Nouns that stand in front of other nouns

6A • Sort — a thing, an activity, or something being put in place?

Three columns.

container • handling • deployment • terminal • scheduling • integration • automation • allocation

6B • Chunk completion

1. Rotterdam is Europe's largest _____. (three words)
2. Berth 4 will be served by _____. (four words)
3. Barges reach the hinterland over _____. (four words)
4. Every movement is booked through _____. (four words)
5. Only two of the four berths have _____. (two words)
6. The eastern quay was built for _____. (three words)
7. Eleven of the two hundred claims concern _____. (four words)
8. Reefer boxes are audited for _____. (two words)

6C • Three that are not interchangeable

a container handling terminal • a terminal operating system • a customs clearance delay

Say which one names a place, which one names software, and which one names an event — and in each case find the head.

6D • The phrasal verbs and the fixed expressions

stack up • take up • set up

f three nouns in a row • f what modifies what • f read it backwards

One of these three fixed expressions is an instruction, one is a warning and one is a question. Match them, and say which you would use first on a strange clause.

6E • Contrast Clinic

Eight items. Pack or unpack each one as the bracket says.

1. They deploy automation on the cranes. (*pack to three words*)
 2. crane automation deployment delays (*unpack, naming two participants*)
 3. The system schedules the allocation of berths. (*pack to three words*)
 4. container handling terminal capacity constraints (*unpack into a sentence with a number*)
 5. Customs clearance was delayed. (*pack to three words*)
 6. terminal operating system integration requirements (*unpack, naming who integrates what*)
 7. The inland waterway network was extended. (*pack to four words*)
 8. inland waterway network provisioning headroom (*unpack into two sentences*)
-

Part 7 • Pronunciation & Fluency Lab ♦ B • THE SYSTEM



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6in}

One phrase, one tune - and where you break it — A pause inside a noun phrase is a bracket the listener can hear.

A long noun phrase is one tune, not eight

7A • Hear 🎧 Track 3.3

the automated gantry crane handover latency REQUIREMENTS — one unit, level, falling on the head the requirements | for handover latency | on automated cranes — three units, three small falls handover latency | shall not exceed | four hundred milliseconds — a sentence, and the number lands

7B • Say

Say all three. Then say this without pausing anywhere:

the automated gantry crane handover latency requirements — eight words, one breath, one fall, and if you pause in the middle your listener will attach the next word to the wrong noun.

7C • Make it mean something

Say these two:

automated / *gantry crane handover* — the pause makes *automated* modify the whole thing
automated gantry crane / *handover* — and now it modifies the crane only

A pause inside a noun phrase is a bracket you can hear, and in a specification read aloud it is often the only thing telling a supplier which reading you meant.

7D • Fluency 30

Thirty seconds: **one eight-word noun phrase said three times, with the break in a different place each time**. Three times, faster.

Part 8 • Writing C • THE PERFORMANCE

A requirement that can only be read one way • 200 words

8A • Read the model

At berth 4, handover latency between the remote console and the crane shall not exceed four hundred milliseconds. [one sentence, one verb, one number: nothing here can be read two ways]

This applies to all four automated rail-mounted gantry cranes, and to no other equipment. [a four-word pre-modified phrase, which is safe because the head is the last word and the scope is stated]

Measurement shall be end to end, from the operator's input to confirmed crane response, in accordance with the test method at annex C. [two post-modifiers doing what six pre-modifiers were doing before, and twice as long]

The twenty per cent design margin specified in clause 3.1 does not apply to this requirement. [a packed noun phrase with a relative post-modifier that anchors it to a clause number – dense, and only one reading survives]

Questions about the writer's choices:

1. The first sentence has a verb and a number. **What would the noun-phrase version have cost?**
2. Paragraph three is longer than the phrase it replaces. **Is that a loss?**
3. The last sentence packs again. **Why is it safe here?**

8B • The anti-model

Automated rail-mounted gantry crane remote operation handover latency performance requirements for berth 4 container handling terminal operations shall be in accordance with applicable system integration provisioning standards.

One sentence, twenty-six words, fourteen of them nouns, no number, and three defensible readings. **Rewrite it as four sentences, each with one number or one reference.**

8C • Plan

the requirement as a sentence with a number → the scope, packed but anchored → the method, in post-modifiers → one packed phrase that is safe because it points at a clause

8D • Write — 200 words

8E • Peer review — three criteria

- ☐ Can a reader find the head of every phrase in one pass? ☐ Does every requirement carry a number or a reference? ☐ Is there any pre-modifier that could attach to two different nouns?

8F • Write it again • 8G • Publish

Rewrite after review. **Publish both, and bracket the head of every phrase over four words.**

Part 9 • The File ♦ A • THE WORLD + C • THE PERFORMANCE

Five phrases a learner unpacks and a field packs
Correct, longer, and instantly identifiable as not written by somebody in the industry.

STUDY FILE - THE PHRASE YOU CAN READ AND CANNOT WRITE

Reading a four-noun phrase is easy. Building one is not, because you cannot hear whether the order is right, so you unpack instead.

WHAT YOU WRITE	WHAT THE FIELD WRITES	SAVED
THE SYSTEM FOR MANAGING CONTAINERS	THE CONTAINER MANAGEMENT SYSTEM	2 WORDS
DELAYS IN THE CLEARANCE OF CUSTOMS	CUSTOMS CLEARANCE DELAYS	3 WORDS
THE TIME TAKEN TO TURN A SHIP ROUND	TURNAROUND TIME	6 WORDS
THE NETWORK OF WATERWAYS INLAND	THE INLAND WATERWAY NETWORK	1 WORD
THE INTEGRITY OF THE COLD CHAIN	COLD-CHAIN INTEGRITY	3 WORDS

HOW TO CHECK ONE YOU HAVE BUILT
read it backwards requirements - latency - handover - crane
find the head the last noun before of, for or at
three at a time four-noun phrases are learned, not derived
the five-second rule if it takes longer to parse, unpack it

Collect five phrases of four words or more from real documents in your field, mark the head of each, and bring one you cannot parse.

Nothing on the left is wrong. All of it is longer

Five phrases, fifteen words saved, same information

Reading backwards is the whole technique

The one you cannot parse is next week's lesson

6in}

Five phrases a learner unpacks and a field packs — Correct, longer, and instantly identifiable as not written by somebody in the industry.

STUDY FILE • The noun phrase you can read and cannot write

9A • Reading it and building it 📶 Track 3.4

Most learners at this level can read a four-noun phrase and almost none will produce one. They unpack instead — *the system for the management of the containers* — which is correct, longer, and instantly identifiable as not written by somebody in the industry.

what you write

the system for the management of the containers

what the field writes

the container management system

why you avoid it

you cannot hear whether the order is

the fix

right
learn the phrase, not the rule: three nouns at a time

9B • Pack three

Take three *of*-phrases you have written in English. **Pack each into a pre-modified noun phrase**, and then read each one backwards to check it.

9C • Role play — what modifies what?

In pairs. One of you reads a four-noun phrase from your own field. The other asks *what modifies what?* **Four rounds each**, and write down any phrase that takes more than five seconds.

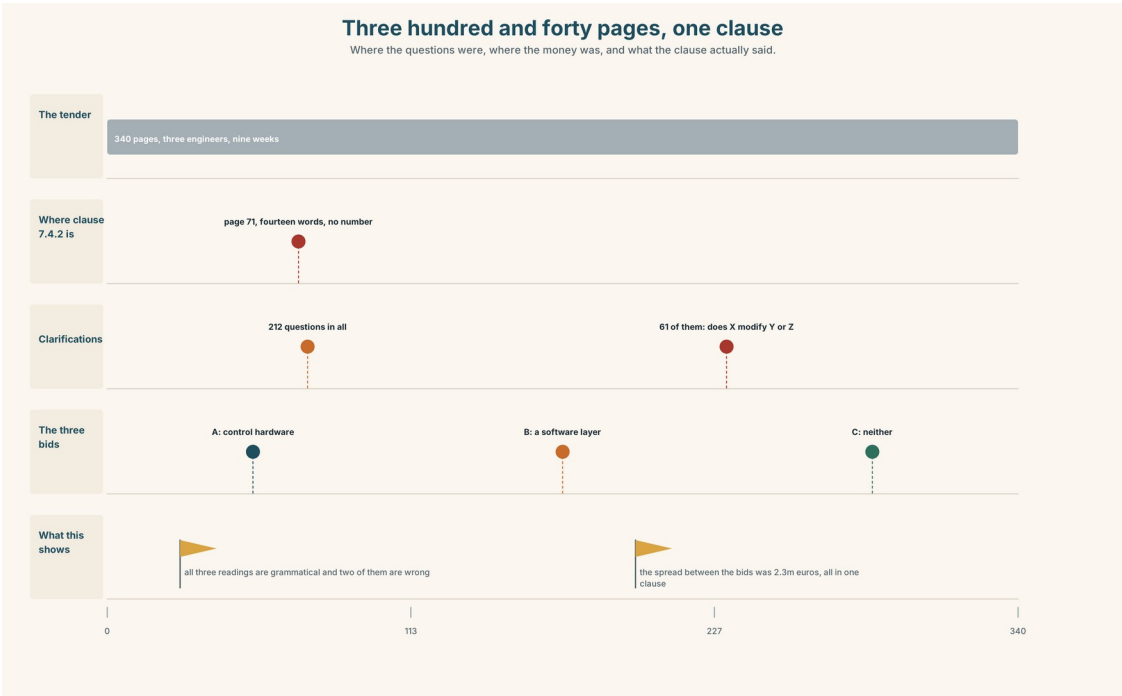
9D • The three you already own

Everybody has three in their own field that they use without thinking. **Name yours**. They are the model for every one you will build.

9E • Transfer — before next week

Collect five noun phrases of four words or more from real documents in your field. Bring them with the head of each one marked, and one you cannot parse.

Part 10 • Mediation & Interaction ♦ C • THE PERFORMANCE



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Three hundred and forty pages, one clause — Where the questions were, where the money was, and what the clause actually said.

10A • Relay

Half the class has clause 7.4.2. Half has the three bid summaries. **Work out where the two point three million euros is, by speaking only.** Ten minutes.

10B • Simplify — for one reader

A supplier writes: *Does "automated" apply to the crane or to the handover?* **Reply in four sentences**, and publish the answer rather than sending it.

10C • Online

Write **one message** proposing that every clarification answer be published to all bidders. **Three sentences**. Say what it costs and what it buys.

10D • Your language

Does your language stack nouns to the left, as English does, or only to the right? **Translate all four versions from Part 2**. Which ones become the same sentence?

Part 11 • The Decision ♦ C • THE PERFORMANCE

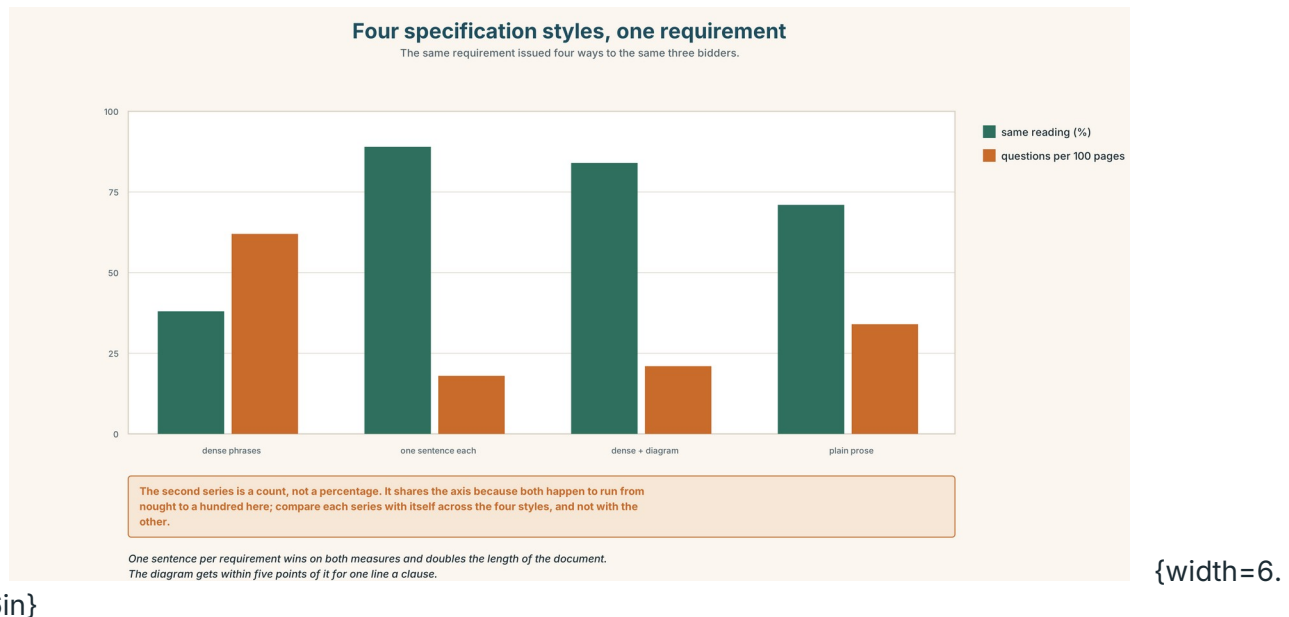
How the port writes a specification

Option one: keep the style. It is dense, it is what the industry writes, and every supplier in this market has read a thousand pages of it.

Option two: every requirement becomes one sentence with a verb and a number. The document doubles in length.

Option three: keep the dense phrases, and require that any noun phrase over six words appear beside a bracket diagram of its head and modifiers, with one requirement, one sentence and one number in the clause itself.

11A • The evidence



6in}

Four specification styles, one requirement — The same requirement issued four ways to the same three bidders.

Track 3.6. Gunnar Holt, bid manager, 30 seconds: "We do not price what you wrote. We price what we think you meant, with a margin for being wrong. On this tender my margin for being wrong was six per cent of the bid, and I will tell you exactly which clause bought it. Make the clause unambiguous and you do not save six per cent — you save it three times, because all three of us stop carrying it."

11B • Talk about it

In threes. Use the frame, and build one noun phrase of five words or more per turn.

I would _____, because _____. The head of that phrase is _____.

11C • Decide

Write **three sentences**: the decision, the evidence, and the clause you would rewrite first.

11D • Model decision

Option three. Option two is right about every individual clause and wrong about the document: a tender that doubles in length is a tender fewer people read to the end, and the clause that loses the argument is then the one nobody reached. The bracket diagram costs a line and does what a pause does in speech. The one-requirement rule is the part that would survive on its own, and the diagram is what makes it checkable before the tender goes out.

11E • The Reveal

What happened. The port adopted the rule. Clarification questions on the next tender fell from two hundred and twelve to seventy-nine, and the proportion about modification fell from twenty-nine per cent to four.

Then something nobody predicted: the spread between the bids narrowed from two point three million euros to three hundred thousand. Nobody had bid differently; they had stopped

pricing three different scopes, and the margin each of them had been carrying for being wrong came out of all three bids at once.

One question. You do not have to answer it out loud.

The port did not negotiate anything. Why did the price fall?

Part 12 • Landing ◆ all three lines

12A • Recycle — fifteen items

1. They deploy automation on the cranes. → _____ (three words)
2. The system schedules berth allocation. → _____ (three words)
3. Customs clearance was delayed. → _____ (three words)
4. Rotterdam is Europe's largest _____ . (three words)
5. Every movement is booked through _____ . (four words)
6. Barges reach the hinterland over _____ . (four words)
7. Only two berths have _____ . (two words)
8. Reefer boxes are audited for _____ . (two words)
9. The terminal is sized for _____ . (two words)
10. Latency is measured _____ . (three words)
11. _____ . (six words — the test to run on a strange clause)
12. _____ . (four words — the end of the argument)
13. The queue will _____ if one crane stops. (two words)
14. Spare capacity is what you _____ in advance. (two words)
15. The new system will _____ in April. (two words)

12B • Consolidate — one task, everything in it

Write 150 words specifying something, with three requirements, each one sentence with one number, and one packed phrase of five words or more with its head marked.

12C • Can-do, with evidence

☐ I can build and parse a complex noun phrase. **Evidence:** 3H, 8D ☐ I can find a head and say what modifies what. **Evidence:** 5E, 6C ☐ I can write a requirement that has one reading. **Evidence:** 10B, 8D ☐ I can chunk a heavy phrase when I say it. **Evidence:** 7B

12D • Replay 🟡

Record 0C again. **Compare.** How many *of*-phrases did you use the first time?

12E • Glossary — fifty items, as a map

HOW MUCH GOES THROUGH · throughput · capacity · utilisation · inventory **WHERE IT STOPS** · bottleneck · latency · headroom **HOW IT IS BUILT** · resilience · interoperability · topology · cadence · provisioning **THE FIVE SPECIFICATION PHRASES** · a single point of failure · the critical path · spare capacity · lead time · a service level **THE NUMBERS IN IT** · peak demand · end to end · scheduled maintenance · turnaround time · design margin **WHAT ENGINEERS DO** · scale back · build in · roll out · stack up · take up · set up **IN THE ROOM** · it depends on the volume · that is the constraint · there is no slack **THE STACKABLE NOUNS** · container · handling · terminal · automation · deployment · integration · allocation · scheduling **THE PHRASES THEY BUILD** · a container handling terminal · an automated stacking crane · the inland waterway network · a terminal operating system · rail-served capacity · deep-

water access · a customs clearance delay · cold-chain integrity **TAKING ONE APART** · three nouns in a row · what modifies what · read it backwards

12F · Next

Unit 4 · How does academic English sound?